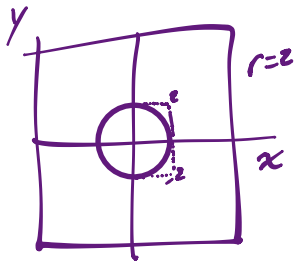
Starting with a circle.

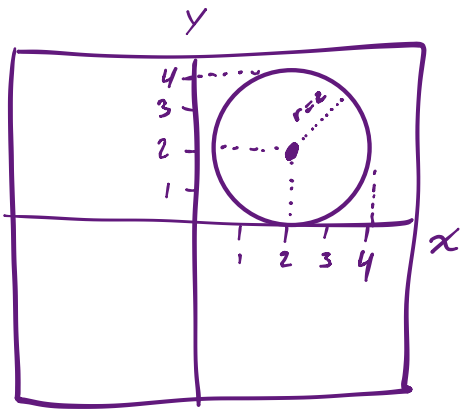
$x^2 + y^2$ = a circle of r^2 Centered @ the origin. → (0,0)?

radius 2

$x^2 + y^2 = 2$ (Centered @ (0,0))



$(x-2)^2 + (y+2)^2 = 2$ gives:



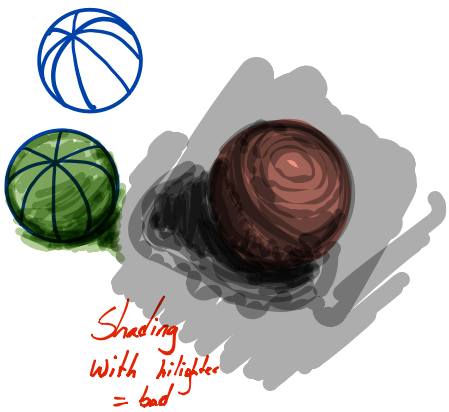
Now to go 3d just add z^2

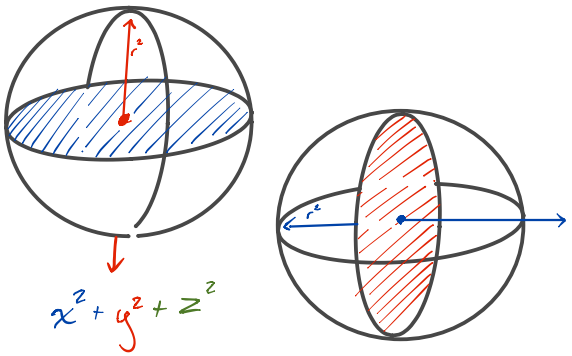
$x^2 + y^2 + z^2 = r^2$

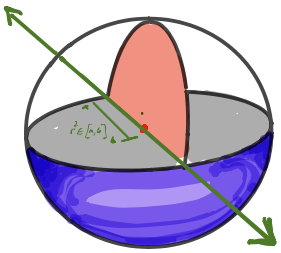
3d sphere Centered @ (0,0,0)

radius = line segment from center to any point along edge.

So if anything ever $= r^2$ then we know it's on the spheres surface.







Visually it now should be obvious why these 3 conditions hold.

Outside Sphere: $x^2 + y^2 + z^2 < r^2$

Inside Sphere: $x^2 + y^2 + z^2 > r^2$

On Surface Sphere: $x^2 + y^2 + z^2 = r^2$

Now to easily control where the sphere is at any point (C_x, C_y, C_z)

→ $((x+C_x)^2 + (y+C_y)^2 + (z+C_z)^2)$

Then lets make (x, y, z) & (C_x, C_y, C_z) all be vectors to more easily control.

Let $P = (x, y, z)$

$C = (C_x, C_y, C_z)$ — where we want to move to. Set the center of the sphere.

Now using dot we can do $(P-C)\cdot(P-C) = \|P-C\|^2$

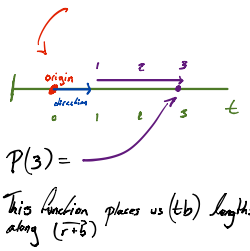
L_2 norm, distance from P to C

Every point P that satisfies $(P-C)\cdot(P-C) = r^2$ is on the surface of the sphere.

Rays. A start + vector

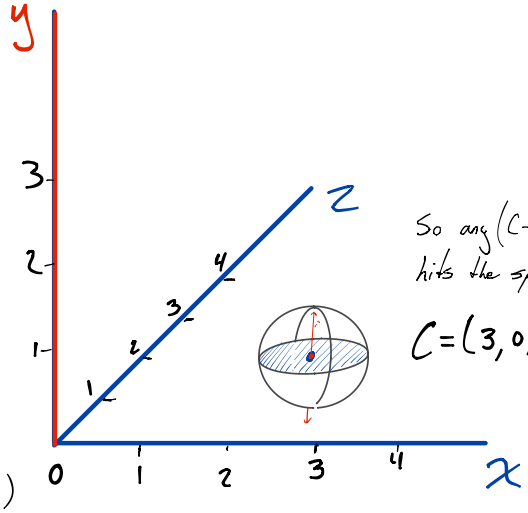
$P(t) = r_{ray\text{-}origin} + t\,d_{direction}$

$P(t) = r + tb$



$P(3) =$

This function places us (tb) lengths along $(\vec{r}+\vec{b})$



So any $(C-P(t)\cdot C-P(t)) = r^2$ hits the sphere.

$C = (3, 0, 2)$

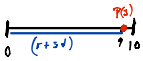
$r = (1,1,1)$ $d = (0,0,3)$

$P(0) = (1,1,1) + 0d = (1,1,1)$

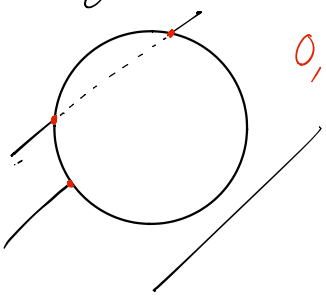
$P(3) = (1,1,1) + 3d$

$= (1,1,1) + (0,0,9)$

$= (1,1,10)$



Using the quadratic formula we can find how many intercepts there are one each ray.



Summary.

$P(x,y,z)$ = hit point

$C(x,y,z)$ = location

$P(t)$ = some point along a vector

$(C-P(t)\cdot C-P(t)) = r^2$ when $P(t)$ hits the sphere.

When doing $t = \frac{-b \pm \sqrt{b^2-4ac}}{2a}$

We will find the descrim will tell us either

$b^2-4ac < 0$ Shot ray miss

$= 0$ one intercept

> 0 two intercepts

$a =$

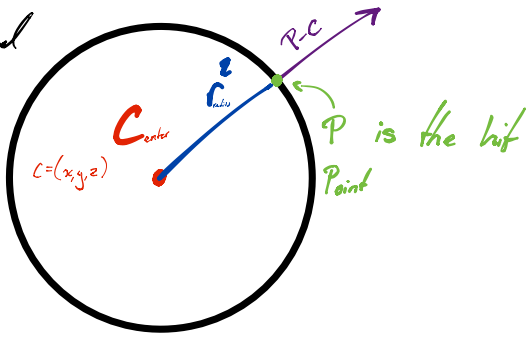
$b =$

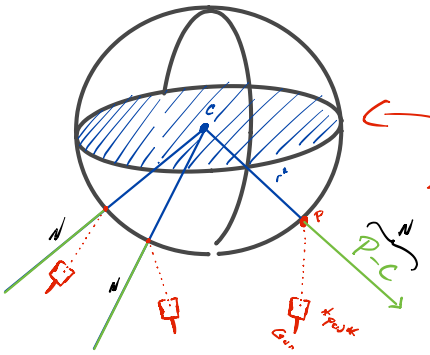
$c =$

Surface Normals

A unit vector perpindicular to the surface at the point of intersection.

Finding a normal for any hit point





lets shoot a ray and see the normal

This is the normal

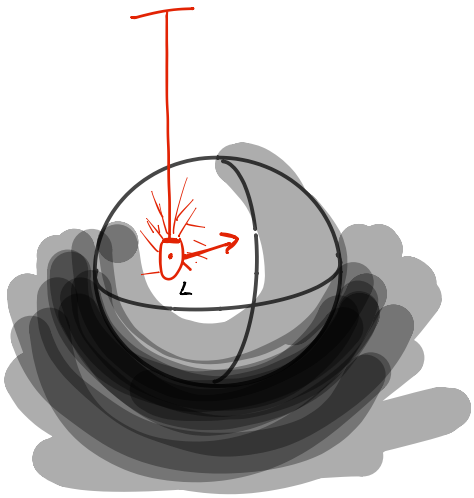
Let $\frac{P-C}{r} = N$

N for normal.

So we have units in a normalized form.

Now from $R_{ay}(t) = O_{rigin} + t\vec{d}$ (Point, vector)

We can do $\vec{N} \cdot \vec{d}$ to find the direction of the normal. Going in or out of the circle.



L is a vector for a ray from the light to the hit point.

Do $\max(0, \vec{L} \cdot \vec{d})$

if 1 it's hitting this lit

Antialiasing

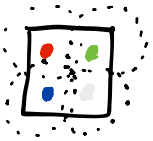
A pixel is not a little square!



This is often how we think of pixels, but in reality we see in $\mathbb{R}$ not $\mathbb{Z}$

Cameras resolution is limited so we capture eventually in $\mathbb{Z}$

In reality a pixel is a combination of area in $\mathbb{R}$

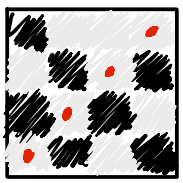


Ok... but how do we anti-alias....?

Right now I am doing whats called "Point sampling".

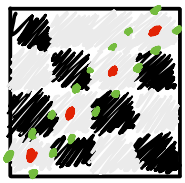
Shooting a single ray and setting a pixel for that first intercept.

The problem is this. How many rays to shoot.



Shoot 4 rays

We only see white



Instead shoot 4 rays for each one, each additional ray in a random direction by an offset of 0.5

Then average between them

